

Review article

Secondary caries risk of different adhesive strategies and restorative materials in permanent teeth: Systematic review and network meta-analysis

Haitham Askar^a, Joachim Krois^b, Gerd Göstemeyer^a, Falk Schwendicke^{b,*}

^a Department of Operative and Preventive Dentistry, Charité – Universitätsmedizin Berlin, Aßmannshäuser Str. 4-6, 14197 Berlin, Germany

^b Department of Oral Diagnostics, Digital Health and Health Services Research, Charité – Universitätsmedizin Berlin, Aßmannshäuser Str. 4-6, 14197 Berlin, Germany

ARTICLE INFO

Keywords:

Detection
Secondary caries
Evidence-based medicine
Restorations
Systematic review

ABSTRACT

Objectives: Secondary caries is a major long-term complication of dental restorations. Different adhesive strategies and restorative materials may affect secondary caries risk. We aimed to systematically review and synthesize the secondary caries risk of different adhesive strategies and restorative materials.

Sources: Medline via PubMed 01/2005-10/2019.

Study selection: Randomized controlled studies with minimum 2 years follow-up, comparing different adhesive strategies and/or restorative materials in permanent teeth were included. Our outcome was the occurrence of secondary caries. Bayesian pairwise and network-meta-analysis were conducted.

Data: We included 50 trials; 19 assessing secondary caries depending on different adhesive strategies, 31 on restorative materials. Studies were published between 2005 and 2017, largely of unclear risk of bias, and included a mean of 40 (range: 8–90) participants and 46 (range: 14–200) placed restorations. Mean follow-up was 43 (range: 24–180) months. Secondary caries was a rare event; the majority of studies did not find any lesions. Network meta-analysis found great uncertainty. 3-step etch-and-rinse adhesives showed the lowest risk of secondary caries, 2-step etch-and-rinse the highest. For restorative materials, resin-modified glass ionomer showed the lowest risk of secondary caries. Most resin composites showed similar risks.

Conclusion: Data from randomized trials comparing different adhesive strategies or restorative materials are extremely scarce. The differences between materials were limited over the observational period of the included studies. The yielded rankings should be interpreted with caution.

Clinical significance: Despite often claimed to be a major complication of restorations, there is surprisingly little data on secondary caries from randomized trials. Longer-term studies may be needed to identify differences in secondary caries risk between materials.

1. Introduction

Secondary caries refers to caries lesions occurring adjacent to an existing restoration (CAR). This might occur due to improper caries management on the patient level, or because of defective restoration margins allowing acidic fluid ingress into the tooth-restoration interzone [1–3]. Secondary caries is often named as major long-term reason for restoration failure, especially for resin-based materials [4–6]; replacement or repair of dental restorations due to secondary caries is a common procedure in daily care.

While in vitro studies are able to compare different restorative materials or strategies (e.g. adhesives, resin composites or other material

classes like glass ionomers) for their impact on secondary caries, using a range of models [3], the findings of such in vitro studies are oftentimes not necessarily coherent between studies or correlating with clinical data [7,8]. Hence, data from clinical trials would be ideal to determine if different restorative materials or strategies come with different risks of secondary caries.

Although many randomized clinical trials have evaluated the clinical performance of different dental materials, and a range of reviews have systematically appraised that performance using pairwise or network meta-analysis [4,9–11], there is no systematic compilation of data on secondary caries. Therefore, the aim of the present study was to systematically review the risk of secondary caries adjacent to different

* Corresponding author at: Charité Centre for Dental Medicine, Department of Oral Diagnosis, Digital Health and Health Services Research, Aßmannshäuser Str. 4-6, 14197 Berlin, Germany.

E-mail address: falk.schwendicke@charite.de (F. Schwendicke).

<https://doi.org/10.1016/j.jdent.2020.103541>

Received 29 August 2020; Received in revised form 25 November 2020; Accepted 26 November 2020

Available online 28 November 2020

0300-5712/© 2020 Elsevier Ltd. All rights reserved.

adhesive systems and restoration materials in permanent teeth, and to perform pairwise and network meta-analysis to synthesize risks of failures.

2. Methods

Reporting of this review follows the PRISMA-NMA guidelines [12, 13].

2.1. Eligibility criteria

We included (1) randomized controlled trials, (2) comparing at least two different adhesive strategies, restorative materials, or their combinations in permanent teeth, (3) with a follow-up period of at least two years (as we did not expect secondary caries to occur in trials with shorter follow-up periods), (4) and placement of the restoration for the treatment of primary caries, replacement of a defective restoration, or non-caries cervical lesions (NCCL). If the same trial was reported in multiple follow-ups, only the most recent publication (longest follow-up) was included.

We excluded (1) retrospective or observational, non-controlled studies, (2) performed on primary teeth, (3) or evaluating restoration repairs, indirect restorations or outdated materials which are not available currently on the market. To restrict the review to materials available and relevant today, the search period was limited from 2005 onwards (see below).

Notably, such a review may not yield a significant number of studies involving one of the most commonly used materials, dental amalgam. However, with the UNEP-led international decision to phase-down and possibly even phase-out dental amalgam in the future, as well as amalgam now being only a relatively small proportion of daily restorative care in many countries, but also accounting for the fact that amalgam has not been compared against most non-amalgam, contemporary materials, it was decided to accept this caveat. It should be highlighted that data comparing amalgam with alternatives, mainly resin composites, has shown amalgam to yield lower risks of secondary caries [14].

2.2. Information sources and search

We searched Medline via PubMed (from 2005, as discussed, until October 2019). The search terms and strategy are displayed in Fig. 1. The reference lists of the included studies were screened to identify possible relevant studies (cross-referencing).

2.3. Study selection and data collection

Two independent researchers performed the electronic research and screened titles and abstracts. Data extraction as well as the risk of bias assessment was performed by one of the authors and verified by a second author. Disagreements between the evaluators were resolved through discussion. A pilot-tested spreadsheet was used for data extraction and management.

2.4. Outcome

Our outcome was the occurrence of secondary caries, assessed via either visual-tactile method, radiographs, or other detection methods. Secondary caries could have been evaluated explicitly or as part of a criteria set like FDI or United States public health service (USPHS) criteria.

2.5. Data items and categorization

The following data were collected: study, year, participants information (age, gender, sample size), adhesive strategies or restorative materials tested (commercial name and details, see below), restorations placed (number, involved surfaces, reasons for restoration [caries, replacement of defective restoration or NCCL]), follow-up (duration, intervals, method) and number of detected secondary caries lesions in each follow-up interval.

For data analysis, the restorative strategies and materials were analyzed in two groups, each employing the following categorization:

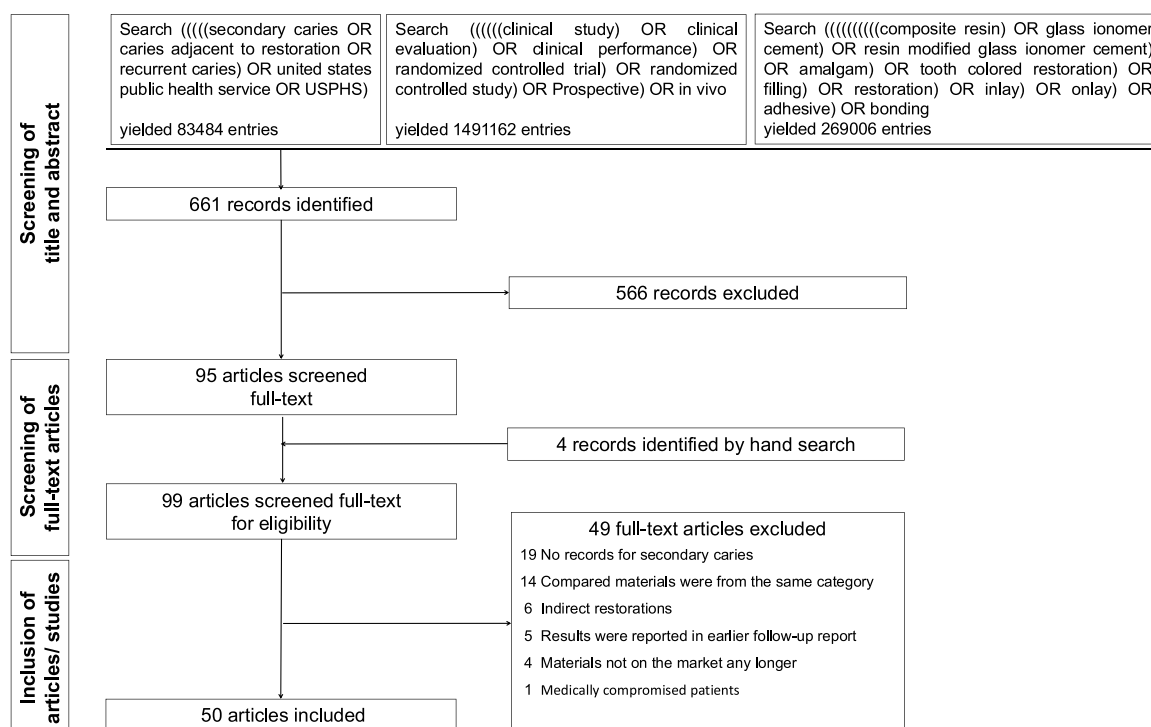


Fig. 1. Flowchart of the review process.

- 1 For adhesive strategies, four groups were used: 2-step etch-and-rinse (2-ER), 3-step etch-and-rinse (3-ER), 1-step self-etch (1-SE), 2-step self-etch (2-SE). In this analysis, only resin-based materials, where any adhesive is used, were compared.
- 2 For restorative materials, the following groups were used: Glass-ionomer cement (GIC), resin-modified glass-ionomer cement (RMGIC), compomer (C, polyacrylic acid-modified composite). Resin composite materials were further sub-classified according to the matrix, filler particles and application mode [15]: nanofilled (N), nanohybrid (NH), microfilled (M), microhybrid (MH), bulkfill (BF), silorane-based (Si). In this analysis, all materials were compared.

Details on the classification of different products involved are presented in appendix Tables S1 and S2, respectively. This categorization system has, with some modifications, been used before [9,15,16].

2.6. Risk of bias

Selection bias (random sequence generation, allocation concealment), detection bias (blinding of participants and personnel, blinding of outcome assessment) and reporting bias (selective reporting, incomplete outcome data) were recorded, and assessed according to the 2011 Cochrane risk of bias tool for randomized clinical trials [17]. Note that a revised version (RoB 2.0) is available by now.

2.7. Synthesis

Bayesian random-effects pairwise and network meta-analyses (NMA) were performed, yielding pairwise, indirect and network estimates, one on adhesive strategies and a separate one on restorative materials. Studies which had compared combinations of adhesive strategy and restorative material, but did not allow to separate the effects of these two factors, were not entered into the analysis. There was no missing data.

Pairwise, indirect and network (mixed comparisons) estimates were made using Bayesian random-effects modelling and Markov Chain Monte Carlo (MCMC) simulations using JAGS implemented in the R package *gemtc* 0.8–2 [18]. Networks of interventions were constructed by plotting different treatments (as nodes) and comparisons (as edges) [19,20]. Binomial likelihood distribution was used for the data and normal distribution for the random effects [21]. We employed an empirically-based prior on τ^2 (between-trial variance) for our base-case analysis based on Turner et al. 2015 [22], assuming our outcomes to be biological and the interventions both being non-pharmaceutical ($\text{LN}(-1.33, 1.45)$). In a sensitivity analysis, a half-normal prior with a variance of 0.5 on τ was used [23].

The parameter range for each network was obtained from data using the *gemtc* package. The first 10,000 MCMC iterations were discarded as “burn-in” and then further 100,000 iterations were undertaken for 4 chains with a thinning of 5 after assessment of the autocorrelation plot. The convergence was assessed based on the Brooks-Gelman-Rubin criteria [24] and inspection of trace plots.

The summary measures were OR and their 95 % credible intervals (95 % CrI) were. Credible intervals are the range of values between the 2.5 % and 97.5 % percentile for a specific parameter; the 95 % CrI reflects that there is a 95 % probability for the parameter to be found within this interval. Further, mean ranks and the cumulative ranking curve (SUCRA) [25] were used to rank groups (adhesive strategies, restorative materials). Rankings and SUCRA-values were graphically displayed [26] and pairwise OR emanating from NMA tabularized.

Heterogeneity was assessed using τ^2 , which captures the absolute extent of statistical heterogeneity; τ^2 smaller than the median of the prior distribution implies low statistical heterogeneity, τ^2 larger than the third quartile of the prior distribution implies substantial statistical heterogeneity; otherwise, moderate statistical heterogeneity was assumed. Inconsistency was assessed via node split. Comparison

adjusted funnel plot were assessed to check for the existence of small study bias [27].

3. Results

3.1. Results of the search and included studies

From 661 identified studies, 50 were included (Fig. 1). The included studies are described in more detail in appendix Tables S3 (adhesive strategies) and S4 (restorative materials). The excluded studies, with reasons for exclusion, are shown in appendix Table S5.

Nineteen studies assessed secondary caries risk depending on different adhesive strategies, 31 on restorative materials. The studies were published between 2005 and 2017 and included a mean of 40 (range: 8–90) participants and 46 (range: 14–200) placed restorations. Mean follow-up was 43 (range: 24–180) months. Generally, secondary carious lesion occurrence was a rare event; the majority of studies did not find any lesions, which may be grounded in the shorter follow-up period. Pooling the available studies revealed two networks, one on adhesive strategies (Fig. 2a) and one on restorative materials (Fig. 2b). The included studies were largely of unclear or high risk of bias (Tables S6, 7). There was some indication for small study bias when inspecting funnel plots for the analysis on adhesive strategies, but not for restorative materials (appendix Fig. S1 and S2).

3.2. Results of synthesis

For adhesive strategies, most comparisons were made between 1- and 2-step self-etch adhesives or 2-step etch-and-rinse adhesives (Fig. 2a). When ranking probabilities and the resulting SUCRA-plots were evaluated (Fig. 3), there was some probability that 3-step etch-and-rinse adhesives were associated with a lower risk of secondary caries (not one secondary lesion was detected in the studies including that strategy), followed by 1-step self-etch and 2-step etch-and-rinse adhesives. 2-step self-etch adhesives showed the highest risk, while the differences generally were limited (Table 1). However, this hierarchy of the interventions is accompanied by substantial uncertainty. The comparison of τ^2 (0.71) with the median (0.54) and the third quartile (0.89) of the prior distribution implied moderate statistical heterogeneity. Node splitting did not suggest the presence of statistically significant inconsistency between direct and indirect evidence in the investigated nodes ($p > 0.05$). Further, the sensitivity analysis using an alternative prior did not yield different rankings or conclusions.

For restoration materials, a less densely populated network with more groups emerged, with some comparators (namely GIs) being only sparsely tested and loosely connected to the network. The most frequently considered materials were microhybrid, nanofilled or nanohybrid resin composites (Fig. 2b). Again, only few secondary lesions developed in the included trials. In ranking probabilities and SUCRA-values (Fig. 3), RMGIs were ranked highest, followed by polyacrylic acid-modified composites (compomers) and nanohybrid resin composites. Nanofilled, microhybrid, microfilled and bulkfill resin composites and GI performed similarly. Silorane-based materials showed the lowest ranks and SUCRA-values, indicating the highest likelihood of secondary caries development according to the included trials (Table 2). Again, the hierarchy of the interventions is accompanied by substantial uncertainty. The comparison of τ^2 (0.68) with the median (0.54) and the third quartile (0.88) of the prior distribution implied moderate statistical heterogeneity. Node splitting found significant inconsistency between direct and indirect estimates for the comparison between nanofilled and microhybrid composites ($p = 0.03$) and nanofilled and bulkfilled composites ($p < 0.05$). The sensitivity analysis using an alternative prior did not yield different rankings or conclusions.

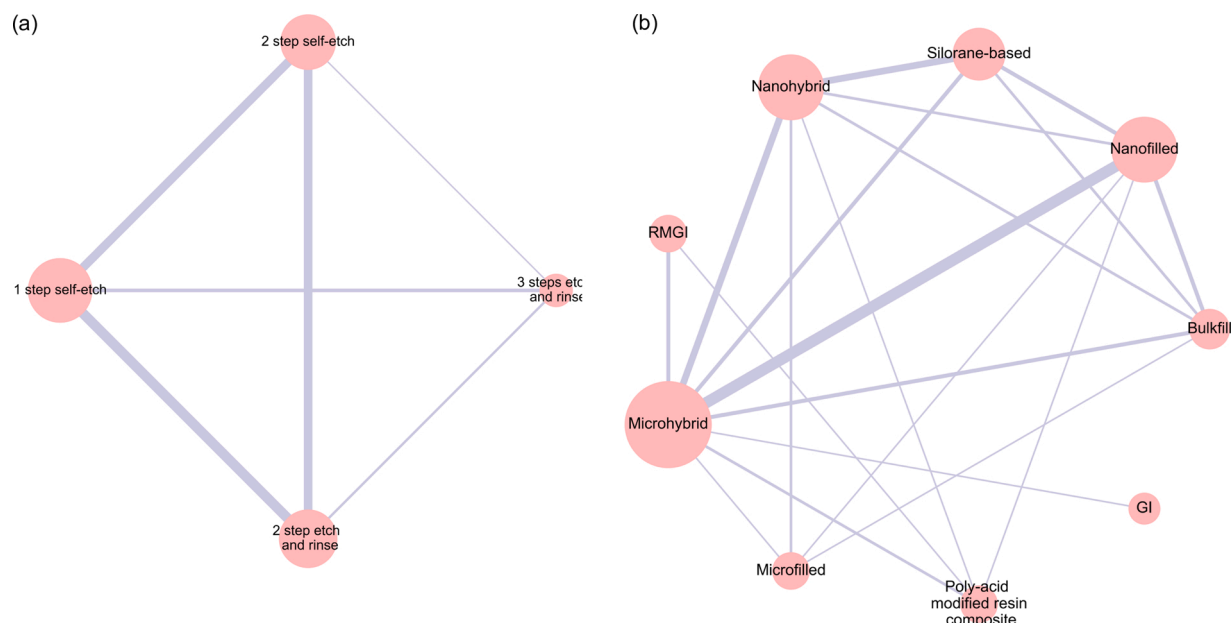


Fig. 2. Network of compared adhesive strategies (a) and restorative materials (b). Nodes (red circles) and connections (blue lines) are used to display actually made pairwise comparisons. The size of the node indicates the number of studies testing this strategy; thickness of the lines indicates the number of studies in which exact comparison was made. RMGI: Resin-modified glass ionomer, GI glass ionomer. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

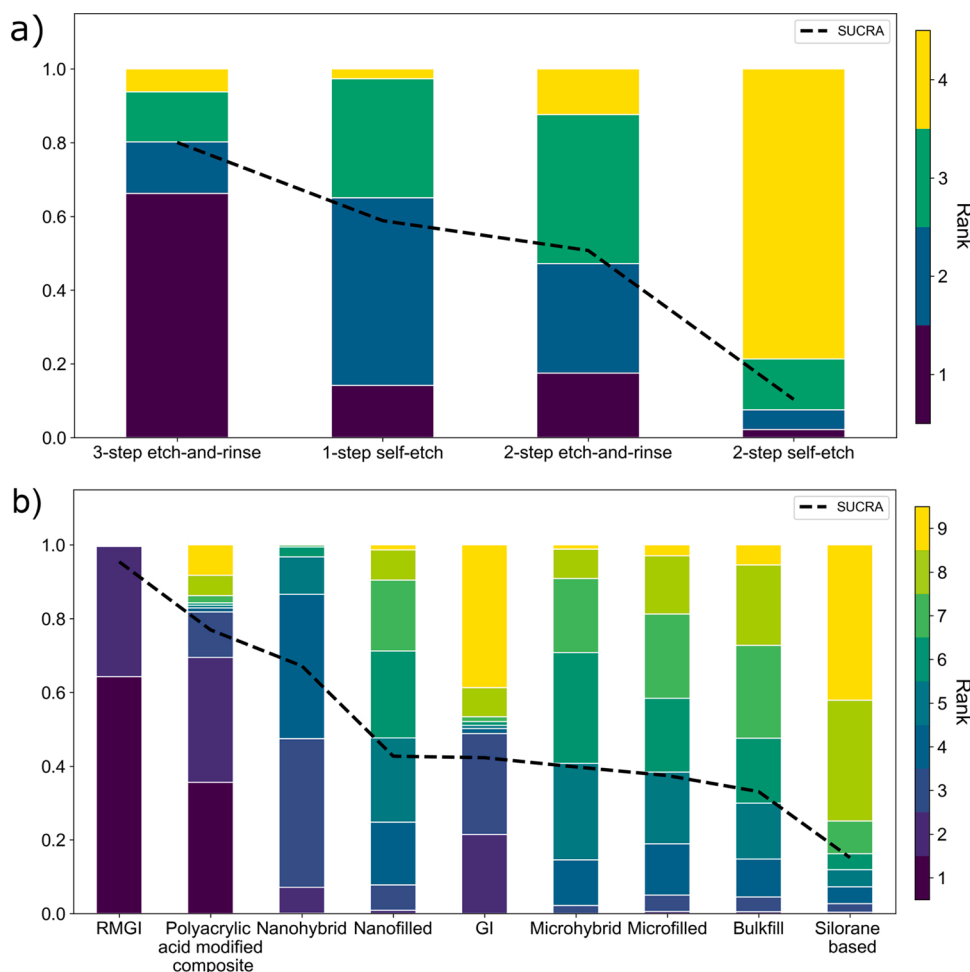


Fig. 3. Cumulative ranks and SUCRA-values. Each adhesive strategy (a) or restorative material (b) is represented by a unique bar. In the present study and using Bayesian network meta-analysis, the ranking of each strategy or material was estimated over a total of 100,000 iterations (calculation rounds). For each iteration, each strategy received a rank; ranks are color-coded (dark blue: first rank, yellow: last rank). Each strategy hence received particular ranks in a certain number of iterations; the resulting frequency (in %) is displayed. A strategy can, for example, be ranked first in ca. 80 % of these iterations (e.g. 3-step etch-and-rinse) indicating this strategy being likely ranked in the top tier, or can be ranked very differently in different iterations (e.g. 2-step etch-and-rinse), with the bar having a wide range of colors, indicating large uncertainty. The strategies are ordered by their SUCRA-value (dotted line), SUCRA-values indicate the cumulative probability of being ranked first, second etc; high SUCRA-value indicate a low risk of secondary caries lesions. RMGI: Resin-modified glass ionomer, GI glass ionomer. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

Table 1

Comparative risks of secondary caries when using different adhesive strategies. Odds Ratio and 95 % credible intervals (95 % CrI) are shown. The risk of failure of the strategy in each row is compared against the ones in each column (e.g. 2-step etch-and-rinse adhesives had a 3.2-times (95 % CrI: 0.073-17.0) increased risk of failure compared with 3-step etch-and-rinse).

	3-step etch-and-rinse	2-step etch-and-rinse	2-step self-etch	1-step self-etch
3-step etch-and-rinse		0.83 (0.001, 8.7)	0.07 (0.001, 3.1)	0.38 (0.01, 7.5)
2-step etch-and-rinse	3.2 (0.073, 17.0)		0.22 (0.004, 3.5)	1.2 (0.11, 7.2)
2-step self-etch	15 (0.32, 190)	4.6 (0.29, 26.0)		5.3 (0.56, 13.0)
1-step self-etch	2.6 (0.13, 10.0)	0.83 (0.01, 14.0)	0.19 (0.004, 1.8)	

4. Discussion

Secondary caries is claimed as a major reason for restoration failure. In the present study, we systematically appraised and synthesized data on secondary caries adjacent to different restoration materials or emanating from different adhesive strategies. We identified 50 randomized controlled trials, most of them employing a small sample of patients and restorations, with high or unclear risk of bias, and restorations followed only over 3–4 years. Given that within these trials and over this observational period, the overall number of events was low, we did not find stark differences between adhesive strategies or restoration materials. Given the sparsely populated network especially on restoration materials, it is difficult to provide firm directions, while the used Bayesian synthesis reflects on this appropriately [23,28]. Nevertheless, any rankings etc. should be interpreted with caution [25,29].

A range of aspects need to be discussed. First, our findings on adhesive systems are consistent with *in vitro* data, with 3-step etch-and-rinse systems performing superior compared to self-etch systems regarding the quality of restoration margins [30]. This is mostly attributed to a less effective enamel surface conditioning of the mild

acids contained in self-etch systems. As margin gaps are considered as a risk factor for secondary caries formation [31], a tight seal of the margins, which can be attained using effective adhesive systems, might reduce secondary caries formation. However, contrary to our findings, clinical data have also shown that 2-step self-etch adhesives have a superior effectiveness in bonding performance compared to 2-step etch-and-rinse and 1-step self-etch systems [11]. Again, it is highlighted that our findings are based on only few events; overall, all adhesives seem to be able to prevent secondary caries development, at least over the limited follow-up period found in the considered trials.

Second, we found RMGICs to have some potential to prevent secondary caries development; again, however, the number of secondary lesions was low even for other materials. Also GIs showed similar performance like the presumably superior composites. GI-based materials are considered as having secondary caries inhibiting properties due to their ability to release fluoride [32]. The difference between RMGICs and GIs may be grounded in the difference in their mechanical properties. Indeed, in direct clinical comparisons, RMGICs do not only perform superior regarding secondary caries formation, but also in marginal gap formation. Different composites came with very similar SUCRA-values, and comparator estimates showed wide credibility intervals, indicating the large uncertainty. From a material perspective, it is conceivable that secondary caries risk is similar, as the differences between the composites are largely in filler type, which is associated more with the physical than the biological or biochemical properties of the composites (e.g. the resin matrix composition or the curing properties are not well reflected by the chosen categorization system, while these two factors could be hypothesized to be more strongly associated with the short-term (curing) and long-term (matrix) margin integrity and hence secondary caries). This was partially reflected by bulk-fill composites performance being at the lower end of materials, possibly as the difference in polymerization shrinking and the limitations in full-depth curing in bulk are likely to be associated with lower margin integrity and hence possibly secondary caries.

Third, and most important, it is clear that over the follow-up period of the included trials, most of which were performed in low-risk patients

Table 2

Comparative risks of secondary caries when using different restoration materials. Odds Ratio and 95 % credible intervals are shown. The risk of failure of the strategy in each row is compared against the ones in each column.

	Nanofilled	Nanohybrid	Microfilled	Microhybrid	Silicone	Glass Ionomer	Bulkfill	Compomer	RMGI
Nanofilled		4.3 (0.35, 94)	0.76 (0.027, 16)	0.9 (0.073, 9.2)	0.0072 (4.2e-11, 10)	0.00013 (5.5e-29, 2.4e+25)	0.57 (0.037, 5.9)	4.3e+09 (1.9e-11, 1.1e+38)	3.8 + 13 (24, 1.0e+40)
Nanohybrid	4.3 (0.35, 94)		0.18 (0.0062, 1.8)	0.22 (0.017, 0.97)	0.0017 (9.9e-12, 1.2)	2.8e-05 (1.8e-29, 5.8e+24)	0.14 (0.0028, 2)	1.0e+09 (5.2e-12, 2.2e+37)	8.1e+12 (6, 2.0e+39)
Microfilled	1.3 (0.062, 37)	5.4 (0.56, 1.6 e+02)		1.2 (0.15, 11)	0.0093 (5.3e-11, 15)	0.00018 (1.1e-28, 4.2e+25)	0.75 (0.015, 30)	6.1e+09 (2.0e-11, 1.6e+38)	5.3e+13 (38, 1.4e+40)
Microhybrid	1.1 (0.11, 14)	4.6 (1.0, 58)	0.86 (0.091, 6.5)		0.0082 (4.3e-11, 8.8)	0.00015 (8.7e-29, 3.2e+25)	0.66 (0.022, 12)	5.5e+09 (2.1e-11, 1.2e+38)	4.3e+13 (36, 1.2e+40)
Silicone	1.4 +e02 (0.095, 2.4e+10)	6 e+02 (0.83, 1.0 e+11)	1.1e+02 (0.067, 1.9 e+10)	1.2e02 (0.11, 2.3e+10)		0.046 (6.9e-27, 7.5e+27)	82 (0.038, 1.8e+10)	1.7e+12 (6.7e-09, 6.2e+42)	2.7e+17 (2.5e+02, 5.7e+41)
Glass Ionomer	7.9e+03 (4.1e-26, 1.8e+28)	3.5 e+04 (1.7 e-25, 5.5 e+28)	5.5 e+03 (2.4 e-26, 9.0 e+27)	6.5e+03 (3.1e-26, 1.2e+28)	22 (1.3e-28, 1.4e+26)		4.2e+03 (2.4e-26, 8.2e+27)	8.0e+14 (2.4e-26, 3.5e+47)	1.9e+18 (5.5e+02, 2.7e+45)
Bulkfill	1.8 (0.17, 27)	7.2 (0.50, 3.6 e+02)	1.3 (0.34, 65)	1.5 (0.081, 45)	0.012(5.5e-11, 27)	0.00024 (1.2e-28, 4.2e+25)		7.5e+09 (3.4e-11, 1.8e+38)	7.3e+13 (42, 1.9e+40)
Compomer	2.3e-10 (8.9e-39, 5.2e+10)	9.9 e-10 (4.5 e-38, 1.9 e+11)	1.6 e-10 (6.3 e-39, 5.1 e+10)	1.8e-10 (8.4e-39, 4.7e+10)	5.8e-13 (1.6e-43, 5.5e+08)	1.2e-15 (2.9e-48, 4.2e+25)	1.3e-10 (5.5e-39, 3e+10)		1.4e+05 (4.2e-27, 3.0e+39)
RMGI	2.6 + 14 (1e-40, 0.041)	1.2 e-13 (4.9. e-40, 0.17)	1.9 e-14 (7.2e-41, 0.026)	2.3e-14 (8.4e-41, 0.028)	3.7e-18 (1.7e-42, 0.0040)	5.2e-19 (3.7e-46, 0.0018)	1.4e-14 (5.3e41, 0.027)	7.1e-06 (3.3e-40, 2.4e+26)	

under controlled (university) settings, only a few lesions developed. Longer-term, practice-based studies found higher incidence in secondary caries [6,33], especially in the later stages of a restoration's lifetime. Moreover, the very few numbers of events might be further confounded with a range of further aspects, like operator skills or technical factors including enamel beveling or the use of rubber dam isolation [34], which were not uniformly applied in all studies, leading to limited transitivity. Hence, our analysis rather serves to confirm that most of the employed restorative strategies and materials performed rather well; the annual rate of secondary caries was low throughout the studies and regardless of the groups assessed. The high incidence rates found in less controlled long-term studies may be ascribed to the fact that secondary caries occurs rather late in the lifespan of a restoration, but may be aggravated by current detection methods not being sufficiently accurate to detect secondary lesions, with over-detection being a frequent problem for many rather sensitive detection methods.

This study has a number of strengths and limitations. First, it used a systematic and reproducible approach, strengthening the robustness of findings. Also, we included only randomized controlled trials, with high internal validity. Admittedly, and as a limitation, transitivity between different studies cannot necessarily be expected (i.e. patients' characteristics, operator skills, risk factors, settings, clinical indications will differ and indirect comparisons may be biased to some degree by that; note however that given that nearly all studies had very few events, the possible distortion from lacking transitivity is likely low). We could also not adjust for possible intransitivity in the treatment effects via meta-regression analysis due to the limited number of trials in the majority of the observed comparisons. Especially the fact that load-bearing and cervical restorations were pooled needs highlighting, while we also want to emphasize the absence of statistical heterogeneity, mainly given the uniformly low number of events. Second, our study used Bayesian network meta-analysis accounting for the various different comparators and introducing uncertainty into our analysis; this allows to reflect the limitations in data quantity. Specifically, it avoids implausible confidence intervals and spuriously zero statistical heterogeneity and does not require to impose approximate normality on binary data, which has been criticised for providing both biased and spuriously precise summary treatment effects as well as inaccurate and imprecise heterogeneity estimates [35]. Also, displaying a range of outcome measures (i.e. rankings, SUCRA-values, pairwise estimates) allows to gauge uncertainty comprehensively. Third, and as a limitation, the data supporting this analysis were sparse; findings may change as new data emerge. The lack of events is likely grounded in the short follow-up time of most trials and the controlled environment; however, and as discussed, it also shows that at least over the first years in the lifetime of a restoration, secondary caries is not a highly relevant problem at least when restorations are placed under controlled conditions and according to standards and manufacturers' instructions. Last, we want to highlight that the chosen classification of materials is only one of many possible. A recent study found that for most materials, similar conclusions will emerge regardless which classification is employed in a meta-analysis [15]. It is also emphasized that particular properties of restorative materials, e.g. buffering capacity, have not been addressed in this review, but have been found potentially relevant for secondary caries development [36].

In conclusion, and based on fifty randomized controlled trials included in this systematic review and meta-analysis, secondary caries was a rather infrequent event during the 3–4 years follow-up period. Moreover, the difference in secondary caries risk between restoration strategies and materials were limited. There is some indication that 3-step etch-and-rinse adhesives are superior to especially 2-step etch-and-rinse ones for preventing secondary caries, but even this conclusion is built on limited data. As long as established materials are used explicitly according to manufacturers' instructions and within the moderately long observational period of the included studies, the risk of secondary caries is low regardless of which material is used.

Declaration of Competing Interest

There are no conflicts to declare.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jdent.2020.103541>.

References

- [1] E.A. Kidd, Diagnosis of secondary caries, *J. Dent. Educ.* 65 (10) (2001) 997–1000.
- [2] F. Schwendicke, M. Kern, U. Blunck, C. Dorfer, J. Drenck, S. Paris, Marginal integrity and secondary caries of selectively excavated teeth in vitro, *J. Dent.* 42 (10) (2014) 1261–1268.
- [3] J.L. Ferracane, Models of caries formation around dental composite restorations, *J. Dent. Res.* 96 (4) (2017) 364–371.
- [4] F. Beck, S. Lettner, A. Graf, B. Bitriol, N. Dumitrescu, P. Bauer, A. Moritz, A. Schedle, Survival of direct resin restorations in posterior teeth within a 19-year period (1996–2015): a meta-analysis of prospective studies, *Dent. Mater.* 31 (8) (2015) 958–985.
- [5] F.F. Demarco, M.B. Correa, M.S. Cenci, R.R. Moraes, N.J. Opdam, Longevity of posterior composite restorations: not only a matter of materials, *Dent. Mater.* 28 (1) (2012) 87–101.
- [6] N.J. Opdam, E.M. Bronkhorst, B.A. Loomans, M.C. Huysmans, 12-year survival of composite vs. amalgam restorations, *J. Dent. Res.* 89 (10) (2010) 1063–1067.
- [7] H. Askar, Y.K. Tu, S. Paris, Y.C. Yeh, F. Schwendicke, Risk of caries adjacent to different restoration materials: systematic review of in situ studies, *J. Dent.* 56 (2017) 1–10.
- [8] H. Askar, J. Krois, G. Göstemeyer, P. Bottenberg, D. Zero, A. Banerjee, F. Schwendicke, Secondary caries: what is it, and how it can be controlled, detected, and managed? *Clin. Oral Investig.* 24 (5) (2020) 1869–1876.
- [9] F. Schwendicke, G. Göstemeyer, U. Blunck, S. Paris, L.Y. Hsu, Y.K. Tu, Directly placed restorative materials: review and network meta-analysis, *J. Dent. Res.* (2016).
- [10] S.D. Heintze, V. Rousson, R. Hickel, Clinical effectiveness of direct anterior restorations—a meta-analysis, *Dent. Mater.* 31 (5) (2015) 481–495.
- [11] M. Peumans, J. De Munck, A. Mine, B. Van Meerbeek, Clinical effectiveness of contemporary adhesives for the restoration of non-carious cervical lesions. A systematic review, *Dental Mater.* 30 (10) (2014) 1089–1103.
- [12] B. Hutton, G. Salanti, D.M. Caldwell, et al., The prisma extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations, *Ann. Intern. Med.* 162 (11) (2015) 777–784.
- [13] D. Moher, A. Liberati, J. Tetzlaff, Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement, *Ann. Intern. Med.* 151 (2009).
- [14] F. Schwendicke, G. Göstemeyer, M. Stolpe, J. Krois, Amalgam alternatives: cost-effectiveness and value of information analysis, *J. Dent. Res.* (2018), 22034518782671.
- [15] F. Schwendicke, U. Blunck, Y.K. Tu, G. Göstemeyer, Does classification of composites for network meta-analyses lead to erroneous conclusions? *Oper. Dent.* 43 (2) (2018) 213–222.
- [16] F. Schwendicke, U. Blunck, S. Paris, G. Göstemeyer, Choice of comparator in restorative trials: a network analysis, *Dent. Mater.* 31 (12) (2015) 1502–1509.
- [17] J.P.T. Higgins, *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.0.1, The Cochrane Collaboration, 2008. <http://www.cochrane-handbook.org>.
- [18] G. van Valkenhoef, G. Lu, B. de Brock, H. Hillege, A.E. Ades, N.J. Welton, Automating network meta-analysis, *Res. Synth. Methods* 3 (4) (2012) 285–299.
- [19] G. Salanti, J.P. Higgins, A.E. Ades, J.P. Ioannidis, Evaluation of networks of randomized trials, *Stat. Methods Med. Res.* 17 (3) (2008) 279–301.
- [20] G. Salanti, F.K. Kavvoura, J.P. Ioannidis, Exploring the geometry of treatment networks, *Ann. Intern. Med.* 148 (7) (2008) 544–553.
- [21] S. Dias, A.J. Sutton, A.E. Ades, N.J. Welton, Evidence synthesis for decision making 2: a generalized linear modeling framework for pairwise and network meta-analysis of randomized controlled trials, *Med. Decis. Making* 33 (5) (2013) 607–617.
- [22] R.M. Turner, D. Jackson, Y. Wei, S.G. Thompson, J.P. Higgins, Predictive distributions for between-study heterogeneity and simple methods for their application in Bayesian meta-analysis, *Stat. Med.* 34 (6) (2015) 984–998.
- [23] T. Friede, C. Röver, S. Wandel, B. Neuenschwander, Meta-analysis of two studies in the presence of heterogeneity with applications in rare diseases, *Biom. J.* 59 (4) (2017) 658–671.
- [24] S.P. Brooks, A. Gelman, General methods for monitoring convergence of iterative simulations, *J. Comput. Graph. Stat.* 7 (4) (1998) 434–455.
- [25] G. Salanti, A.E. Ades, J.P. Ioannidis, Graphical methods and numerical summaries for presenting results from multiple-treatment meta-analysis: an overview and tutorial, *J. Clin. Epidemiol.* 64 (2) (2011) 163–171.
- [26] B. Rouse, A. Chaimani, T. Li, Network meta-analysis: an introduction for clinicians, *Intern. Emerg. Med.* 12 (1) (2017) 103–111.
- [27] A. Chaimani, J.P. Higgins, D. Mavridis, P. Spyridonos, G. Salanti, Graphical tools for network meta-analysis in STATA, *PLoS One* 8 (10) (2013), e76654.

- [28] T. Stijnen, T.H. Hamza, P. Özdemir, Random effects meta-analysis of event outcome in the framework of the generalized linear mixed model with applications in sparse data, *Stat. Med.* 29 (29) (2010) 3046–3067.
- [29] G. Salanti, C. Del Giovane, A. Chaimani, D.M. Caldwell, J.P. Higgins, Evaluating the quality of evidence from a network meta-analysis, *PLoS One* 9 (7) (2014), e99682.
- [30] R. Frankenberger, F.R. Tay, Self-etch vs etch-and-rinse adhesives: effect of thermo-mechanical fatigue loading on marginal quality of bonded resin composite restorations, *Dent. Mater.* 21 (5) (2005) 397–412.
- [31] H. Askar, F. Brouwer, M. Lehmensiek, S. Paris, F. Schwendicke, The association between loading of restorations and secondary caries lesions is moderated by the restoration material elasticity, *J. Dent.* 58 (2017) 74–79.
- [32] S. Amend, R. Frankenberger, S. Lückner, E. Domann, N. Krämer, Secondary caries formation with a two-species biofilm artificial mouth, *Dent. Mater.* 34 (5) (2018) 786–796.
- [33] M. Laske, N. Opdam, E. Bronkhorst, J. Braspenning, M. Huysmans, Ten-year survival of ClassII restorations placed by general practitioners, *JDR Clin. Trans. Res.* 3 (2016) 292–299.
- [34] J. Ferracane, T. Hilton, A. Korpak, J. Gillette, P.S. McIntyre, J. Berg, P. Northwest, Use of caries prevention services in the Northwest PRECEDENT dental network, *Community Dent. Oral Epidemiol.* 39 (1) (2011) 69–78.
- [35] D. Jackson, I.R. White, When should meta-analysis avoid making hidden normality assumptions? *Biom. J.* 60 (6) (2018) 1040–1058.
- [36] I. Nedeljkovic, J. De Munck, V. Slomka, B. Van Meerbeek, W. Teughels, K.L. Van Landuyt, Lack of buffering by composites promotes shift to more cariogenic bacteria, *J. Dent. Res.* 95 (8) (2016) 875–881.